

Chapter 13- Dynamic Routing & RIP

What is Dynamic Routing?

Dynamic routing is a method where **routers automatically learn routes** and update their routing tables by exchanging information with other routers.

Instead of manually adding routes (like in static routing), routers talk to each other using **routing protocols**.

How Dynamic Routing Works?

Routers use routing protocols like:

- RIP
- EIGRP
- OSPF
- IS-IS
- BGP

These protocols help routers decide:

- ✓ Which path is best
 - ✓ How far the destination is
 - ✓ How fast the link is
 - ✓ What to do if a link fails
-

Why Dynamic Routing is Used?

-  Automatic route updates
-  No need to configure routes manually
-  Best for medium & large networks
-  Automatically adjusts when link goes down

Real-Life Example

Think about **Google Maps** 

- It automatically finds the best route
- If road is blocked → it changes route

Same way, routers automatically find the best path.

Types of Dynamic Routing Protocols

1 Interior Gateway Protocol (IGP)

Used inside the same organization.

Examples:

- RIP
 - EIGRP
 - OSPF
 - IS-IS
-

2 Exterior Gateway Protocol (EGP)

Used between different organizations (like ISPs).

Example:

- BGP
-

Routing Algorithms Explained

1 Distance Vector

Used in:

- RIP
- EIGRP

What is Distance Vector?

- Router only knows:
 - Distance (how far)
 - Direction (which neighbor)

It does NOT know full network map.

Real-Life Example

Imagine asking your friend:
"How many cities to reach Mumbai?"

Friend says: 3 cities.

You trust your friend without knowing full map.

That is Distance Vector.

2 Link State

Used in:

- OSPF
- IS-IS

What is Link State?

- Router builds complete network map
- Knows every link and cost

Real-Life Example

You open Google Maps and see entire city map.

That is Link State.

Path Vector (EGP)

Used in:

- BGP

It keeps full path information of Autonomous Systems.

RIP (Routing Information Protocol)

 **Type:**

Distance Vector

 **Metric:**

Hop Count

 **Maximum Hop:**

15

16 = Unreachable ❌

 **Administrative Distance:**

120

RIP Versions

◆ RIP v1

- Classful
- No subnet mask support
- Broadcast updates

◆ RIP v2

- Classless
- Supports VLSM
- Supports Authentication
- Uses Multicast 224.0.0.9



Administrative Distance (AD Value)

AD = Trust Level of Routing Protocol

Lower value = More trusted

Protocol	AD
----------	----

Directly Connected	0
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Static	1
--------	---

BGP External	20
--------------	----

EIGRP	90
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OSPF	110
------	-----

RIP	120
-----	-----

EIGRP External	170
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BGP Internal	200
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RIP Timers (Very Important)

Timer	Value	Purpose
Update	30 sec	Sends routing updates
Invalid	180 sec	Mark route invalid
Hold-down	180 sec	Wait before accepting changes
Flush	240 sec	Delete route

How RIP Works (Full Flow)

- 1 Router sends routing table every 30 seconds
- 2 Neighbor receives update
- 3 Router calculates hop count
- 4 Routing table updates

More hops = less preferred ❌

Less hops = best route ✅



RIP LAB

- PC1 → 192.168.1.2
- PC2 → 192.168.4.2

🔧 Router Configuration (Step-by-Step)

◆ Router 1 Configuration

Step 1: Enter Privileged Mode

```
Router> en
```

👉 Enables administrative mode.

Step 2: Enter Global Configuration

```
Router# conf t
```

Step 3: Set Hostname

```
hostname R1
```

👉 Makes identification easy.

Step 4: Configure Interfaces

```
int g0/0
```

```
ip add 192.168.10.1 255.255.255.0
```

```
no shut
```

👉 Connects to R2

```
int g0/1  
ip add 192.168.20.1 255.255.255.0  
no shut
```

👉 Connects to R3

```
int g0/2  
ip add 192.168.1.1 255.255.255.0  
no shut
```

👉 Connects to PC1 network

◆ Router 2 Configuration

```
hostname R2  
int g0/0  
ip add 192.168.10.2 255.255.255.0  
no shut
```

👉 Connects to R1

```
int g0/1  
ip add 192.168.40.1 255.255.255.0  
no shut
```

👉 Connects to R4

◆ Router 3 Configuration

```
hostname R3  
int g0/0  
ip add 192.168.30.1 255.255.255.0  
no shut
```



```
int g0/1
ip add 192.168.20.2 255.255.255.0
no shut
```

◆ Router 4 Configuration

```
hostname R4
int g0/0
ip add 192.168.40.2 255.255.255.0
no shut
int g0/1
ip add 192.168.30.2 255.255.255.0
no shut
int g0/2
ip add 192.168.4.1 255.255.255.0
no shut
```

👉 Connects to PC2



PC Configuration

PC1:

- IP: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.1.1

PC2:

- IP: 192.168.4.2
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.4.1



Enable RIP on All Routers

◆ R1

router rip

version 2

network 192.168.10.0

network 192.168.20.0

network 192.168.1.0

passive-interface g0/2

Why passive-interface g0/2?

- g0/2 connects to PC1
- PC is not a router
- No need to send RIP updates there
- Improves security
- Reduces unnecessary traffic

◆ R2

router rip

version 2

network 192.168.10.0

network 192.168.40.0

◆ R3

router rip

version 2

network 192.168.20.0

network 192.168.30.0

◆ R4

router rip

version 2

network 192.168.30.0

network 192.168.40.0

network 192.168.4.0

passive-interface g0/2

Why passive-interface on R4?

Same reason:

- g0/2 connects to PC2
 - No RIP updates needed toward PC
-

Verification Commands

✓ Check Routing Table

show ip route

- R = RIP route
- C = Connected
- L = Local

Example:

R 192.168.4.0 [120/2]

120 = AD

2 = Hop count

✓ Check Protocol Running

show ip protocols

Shows:

- RIP version
 - Timers
 - Networks added
 - Passive interfaces
-

✓ Check RIP Database

show ip rip database

Shows:

- All learned routes
-

Final Testing

From PC1:

ping 192.168.4.2

If reply comes 

RIP is working perfectly 

Important Points

- RIP is Distance Vector
 - Metric = Hop count
 - Max hop = 15
 - AD = 120
 - RIP v2 supports VLSM
 - Uses multicast 224.0.0.9
 - Timers: 30 / 180 / 180 / 240
-

Final Understanding

Dynamic Routing = Automatic path selection 

RIP = Simple protocol for small networks

Passive Interface = Security + Less traffic



COMPLETE SUMMARY



What is Dynamic Routing?

Dynamic Routing is a routing method where:

- ✓ Routers automatically learn routes
- ✓ Routers exchange information with neighbors
- ✓ Routing tables update automatically
- ✓ No need to manually configure every route



Unlike Static Routing, admin does NOT manually add all routes.



How Dynamic Routing Works?

Routers use **Routing Protocols** like:

- RIP
- EIGRP
- OSPF
- IS-IS
- BGP

These protocols help routers decide:

- ✓ Best path
- ✓ Distance to destination
- ✓ Link speed
- ✓ What to do if link fails



Example: Like Google Maps automatically changing route when traffic appears 🚦

Why Dynamic Routing is Used?

-  Automatic route updates
 -  Best for medium & large networks
 -  Adapts to failures
 -  Saves administrator time
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Types of Dynamic Routing Protocols

1 IGP (Interior Gateway Protocol)

Used **inside** same organization

Examples:

- RIP
 - EIGRP
 - OSPF
 - IS-IS
-

2 EGP (Exterior Gateway Protocol)

Used **between** organizations (ISPs)

Example:

- BGP
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Routing Algorithms

1 Distance Vector

Used in:

- RIP
- EIGRP

Router knows:

- Distance (Hop count)
- Direction (Next hop)

 Does NOT know full network map

 Example: Trusting friend's information without seeing full map

2 Link State

Used in:

- OSPF
- IS-IS

✓ Router builds full network topology

✓ Knows all links and costs

 Example: Seeing full Google Map

3 Path Vector

Used in:

- BGP

✓ Maintains full AS path information

● RIP (Routing Information Protocol)

✚ Basic Details

- Type: Distance Vector
- Metric: Hop Count
- Max Hop: 15
- 16 = Unreachable ✖
- Administrative Distance (AD): 120

✚ RIP Versions

◆ RIP v1

- Classful
- No VLSM
- Broadcast updates

◆ RIP v2

- Classless
 - Supports VLSM
 - Supports Authentication
 - Uses Multicast 224.0.0.9
-

Administrative Distance (AD)

👉 AD = Trust level of routing source

Lower AD = More trusted

Protocol	AD
----------	----

Connected	0
-----------	---

Static	1
--------	---

BGP External	20
--------------	----

EIGRP	90
-------	----

OSPF	110
------	-----

RIP	120
-----	-----

EIGRP External	170
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BGP Internal	200
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RIP Timers (Very Important 🔥)

Timer	Value	Purpose
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Update	30 sec	Send updates
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Invalid	180 sec	Mark route invalid
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Hold-down	180 sec	Prevent quick changes
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Flush	240 sec	Delete route
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How RIP Works (Step-by-Step Flow)

- 1 Sends full routing table every 30 sec
 - 2 Neighbor receives update
 - 3 Adds +1 hop
 - 4 Updates routing table
 - 5 Best route = Least hop count
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Passive Interface

Used when:

- ✓ Interface connects to PC
- ✓ No need to send routing updates
- ✓ Improves security
- ✓ Reduces traffic

Command:

passive-interface g0/2

Verification Commands

Command	Purpose
show ip route	View routing table
show ip protocols	Check RIP status
show ip rip database	View RIP routes

MIND MAP (Detailed Structure)

Dynamic Routing

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|— Definition

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|— Automatic learning

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|— Routing protocols

|

|— No manual routes

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|— Why Used

|

|— Automatic updates

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|— Failure handling

|

|— Scalable

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|— Types

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|— IGP

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|— RIP

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|— EIGRP

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|— OSPF

|

|— IS-IS

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|

|— EGP

|

|— BGP

|

|— Algorithms

|

|— Distance Vector

|

|— Link State

| └─ Path Vector

|

└─ RIP

└─ Distance Vector

└─ Hop Count

└─ Max Hop 15

└─ AD 120

└─ Timers

| └─ 30

| └─ 180

| └─ 180

| └─ 240

└─ RIP v1

└─ RIP v2

Q & A

Q 1 What is Dynamic Routing?

Answer:

Dynamic routing is a routing method where routers automatically exchange routing information and update routing tables using routing protocols.

Q 2 Why use Dynamic Routing instead of Static?

Answer:

Because it:

- Automatically updates routes
 - Handles link failures
 - Scales well
 - Requires less manual configuration
-

Q 3 What are IGP and EGP?

Answer:

IGP is used inside one organization (RIP, OSPF, EIGRP).

EGP is used between organizations (BGP).

Q 4 What is Distance Vector algorithm?

Answer:

It is a routing algorithm where router knows only:

- Distance
- Direction

It does not know full network map.

Q 5 What is Link State algorithm?

Answer:

In Link State, router builds full network topology and calculates shortest path.

Q 6 What is Path Vector?

Answer:

Used in BGP, it maintains full AS path information.

Q 7 What is RIP?

Answer:

RIP is a Distance Vector routing protocol that uses hop count as metric.

Q 8 What is the maximum hop count in RIP?

Answer:

15 hops.

16 means unreachable.

Q 9 What is Administrative Distance?

Answer:

Administrative Distance is the trust level of routing protocol.

Lower AD = More trusted.

Q 10 What is RIP AD value?

Answer:

RIP has AD = 120.

Q 1 1 Difference between RIP v1 and RIP v2?

RIP v1	RIP v2
Classful	Classless
No VLSM	Supports VLSM
Broadcast	Multicast
No Authentication	Supports Authentication

Q 1 2 What multicast address does RIP v2 use?

Answer:
224.0.0.9

Q 1 3 Explain RIP Timers.

Answer:

- Update → 30 sec
 - Invalid → 180 sec
 - Hold-down → 180 sec
 - Flush → 240 sec
-

Q 1 4 What is Passive Interface?

Answer:

It disables routing updates on specific interface but still advertises network.

Used for security and to reduce traffic.

Q 1 5 How to verify RIP?

Answer:

- show ip route
 - show ip protocols
 - show ip rip database
-

Q 1 6 What does R 192.168.4.0 [120/2] mean?

Answer:

R → RIP

120 → AD

2 → Hop count



FINAL CONCLUSION

- 🔥 Dynamic Routing is essential for scalable networks.
 - 🔥 RIP is simple but limited (max 15 hops).
 - 🔥 RIP v2 is better than v1.
 - 🔥 Understanding AD, timers, and algorithms is very important for interviews.
 - 🔥 Passive interface improves security and efficiency.
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